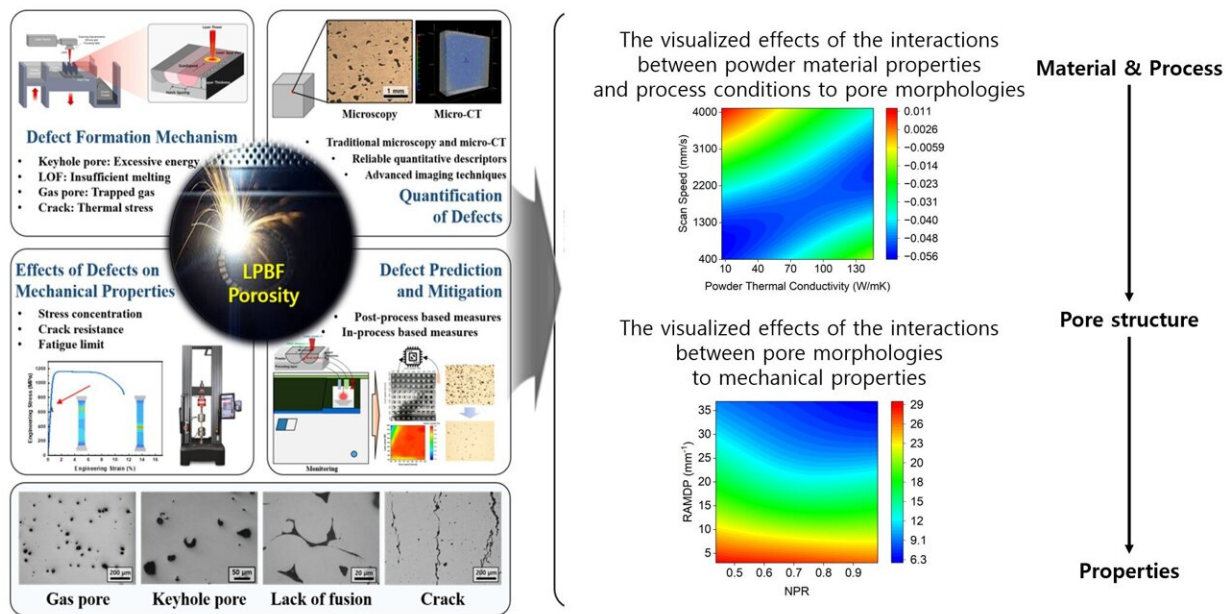


AI enables defect-aware prediction of metal 3D-printed part quality

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Conceptual diagram showing AI-based analysis and prediction of how powder characteristics and process conditions affect defects and component performance in metal 3D printing processes. Credit: Korea Institute of Materials Science (KIMS)

A research team led by Dr. Jeong Min Park of the Nano Materials Research Division at the Korea Institute of Materials Science (KIMS), in collaboration with Dr. Jaemin Wang and Prof. Dierk Raabe of the Max Planck Institute in Germany, has developed an artificial intelligence

(AI)-based model capable of assessing the likelihood and characteristics of internal defects during process design.

This research achievement is expected to significantly enhance the reliability of metal additive manufacturing parts and greatly expand their applicability for mass production in industrial settings. The work is [published](#) in the journal *Acta Materialia*.

Metal additive manufacturing has attracted attention as a next-generation production technology capable of fabricating complex, high-value components. However, its industrial application has been limited due to microscopic internal defects generated during the process, which can lead to component failure and performance degradation.

Conventional quality evaluation has focused primarily on simple indicators such as porosity. In practice, however, the impact on mechanical performance varies significantly depending on the shape, size, location, and distribution of defects.

Moving beyond simple defect measures

To address these challenges, the research team developed an explainable artificial intelligence ([Explainable AI](#)) model capable of systematically analyzing and predicting the relationships among metal additive manufacturing process conditions, defect morphology, and mechanical performance.

This approach enables the prediction of potential internal defects and their impact on performance from the process design stage, presenting a new framework for defect-aware process design and quality management.

How the new AI model works

The core feature of the developed AI model lies in its ability to analyze and predict internal defects generated during the [laser powder bed fusion](#) (LPBF) process of metal additive manufacturing based on morphological characteristics—such as shape and distribution—rather than simply the number or fraction of defects.

By utilizing microstructural images, the model automatically analyzes pore size, non-circularity, and spatial distribution, and directly correlates these factors with mechanical properties, enabling a quantitative explanation of how defects influence actual performance.

In particular, the model is designed to explain why defects increase and performance deteriorates under certain process conditions, distinguishing it from conventional "black-box" AI models whose decision-making processes are not transparent.

Building a broad, data-driven framework

The research team comprehensively analyzed process conditions, [powder characteristics](#), defect images, and mechanical property data across various metal additive manufacturing materials, including steel, aluminum alloys, and titanium alloys, and used these datasets to train the AI model.

Through this approach, they established an integrated framework capable of stepwise prediction—assessing how process variables and powder characteristics influence defect formation, and how defect morphology subsequently affects mechanical performance.

Industrial impact and future directions

This technology can significantly improve the quality reliability of metal 3D-printed components and accelerate their mass production for high-value parts.

In particular, it can be utilized for process optimization and quality control of metal additive manufacturing across industries requiring highly reliable metal components, such as aerospace, defense, and mobility. By reducing defect rates as well as material waste and rework costs, it is expected to enhance overall industrial production efficiency.

Dr. Jeong Min Park of KIMS, the lead inventor, stated, "This research goes beyond simply reducing defects in metal 3D-printed components; it establishes a scientific framework that explains how specific types of defects directly influence performance. We expect this work to contribute to the broader industrial adoption of metal additive manufacturing, particularly in high-performance sectors such as aerospace, space, and defense."

The research team plans to conduct follow-up studies to expand this technology into a [digital twin-based](#) quality management system applicable to industrial settings.

More information: Jaemin Wang et al, Data-Driven analysis relates mechanical properties to pore morphology in laser powder bed fusion, *Acta Materialia* (2026). [DOI: 10.1016/j.actamat.2025.121751](https://doi.org/10.1016/j.actamat.2025.121751)

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